OptSem-C Supplemental Material

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Research objects

This supplement describes the reproducibility material for *OptSem-C: Public Optimizer Contracts for Query Engine Portability.* The artifact contains the grounded public-contract relation, generated query-probe benchmark, executable SQL validation corpus, external optimizer-motif representatives, projection and repair certificates, provenance records, and integrity checks. No private datasets, account credentials, or hidden engine traces are required.

Object	Count	Meaning
Verified public-contract rules	287	Guarded modal rules over optimizer behavior, each linked to public evidence.
Source-linked evidence spans	287	Public URL, line range, source metadata, and segment hash for every grounded rule.
Public source records	26	Documentation and system-description records used by the grounded mainline.
Generated SQL probes	4,216	Executable query probes generated from the optimizer-feature domain and validity constraints.
Valid feature interactions	7,112	Renderable optimizer-feature interactions covered by the probe generator.
External optimizer motifs	90/90	External workload and optimizer motifs covered and executed by representative probes.
Projection-resolution subsets	256	Exhaustive retained-field vocabularies evaluated over the grounded comparison denominator.
Headline false witnesses	498	False portability witnesses across keyword, yes/no, and operator-only projections.
Side-balanced witnesses	498	False witnesses whose left and right signatures are both supported by public source records.

Reproducibility paths

Fast verification. The first command checks package coherence, frozen certificates, paper alignment, source-witness support, and file hygiene.

```
cd artifact
PYTHONDONTWRITEBYTECODE=1 ./run_mainline_checks.sh
```

The expected result is that all fast checks pass and no package-integrity row fails.

Deep replay. The deep replay regenerates derived results from the grounded corpus and benchmark specifications, executes SQL validations and external motif representatives, checks projection and repair certificates, validates PDF/reference/table alignment, and rebuilds manifests.

```
cd artifact
PYTHONDONTWRITEBYTECODE=1 ./run_deep_checks.sh
```

A stricter no-cache replay removes regenerable outputs before replaying the pipeline:

```
cd artifact
PYTHONDONTWRITEBYTECODE=1 ./run_from_scratch_no_cache.sh
```

Certificate families

Family	What is checked
Grounded integrity	Rule counts, provenance coverage, source hashes, state joins, contract maps, and absence of merge conflicts.
Projection kernels	Exact/projection signatures, false-equivalence witnesses, information profiles, mutation projections, and exhaustive retained-field lattices.
Frontier antichains	Minimal safe and maximal unsafe retained-field vocabularies; monotonicity and coverage of the finite repair frontier.
Repair certificates	Minimum separator fields, hitting-set certificates, held-out feature stability, and layer+placement robustness.
Benchmark probes	Feature-domain validity, interaction coverage, SQL rendering, SQL execution, and multi-catalog SQL validation.
External motifs	Coverage and execution of optimizer/workload motifs drawn from recognized benchmark and query-processing families.
Paper alignment	Numeric claims, table sources, references, PDF integrity, manuscript style, and package consistency.
Evidence balance	Source breadth and left/right source support for each headline false-portability witness.
Repository quality	Architecture boundaries, package installability, cleanliness, manifest coherence, and unit tests.

Main reproducibility targets

The key numbers are: keyword projection declares 2,284 equivalences and 254 false equivalences; operator-only projection declares 2,268 equivalences and 238 false equivalences; yes/no projection declares 2,036 equivalences and 6 false equivalences; a strict exact-contract negative control declares 2,030 equivalences and zero false equivalences. The exhaustive field-resolution lattice evaluates all 256 retained-field subsets; 44 are unsafe, and after removing action-variant identifiers the minimum safe semantic vocabularies have size two. The layer+placement basis resolves all 498 headline witnesses. SQL validation executes 12,648 probe-catalog pairs with zero failures across three deterministic catalog densities.

Interpretation boundary

OptSem-C evaluates public optimizer contracts rather than hidden implementation behavior. The corpus therefore proves what public evidence commits an engine to, not what an unobservable optimizer happens to do on a private build. The SQL validation checks that the generated probe corpus is executable and feature-realizing; it is not a latency benchmark. External workload motifs are used as optimizer-feature denominators rather than copied as third-party benchmark scores. These distinctions keep the artifact aligned with the paper's central claim: coarse optimizer comparison fabricates portability when it drops query-processing fields, and the finite repair frontier identifies the missing semantics.